Continuous glucose monitoring use and glucose variability in very young children with type 1 diabetes (VibRate): A multinational prospective observational real-world cohort study

Speaker
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Dovc K, Name MV, Bizjan BJ, et al. Continuous glucose monitoring use and glucose variability in very young children with type 1 diabetes (VibRate): A multinational prospective observational real-world cohort study. *Diabetes Obes Metab.* 2022; 24(3): 564-569.

Background

Type 1 diabetes (T1D) is complicated for very young children for their developmental period.

It is critical to mitigate exposure to hyperglycemia and pronounced glucose fluctuations.

Evidence has shown efficacy of continuous glucose monitoring (CGM) among adults and elder children yet not for very young children.

Study Objective

CGM use in young children with T1D on insulin pump therapy would be associated with diminished glucose excursions and increased time spent in range (TIR).

TIR: 70-180 mg/dl [3.9-10.0 mmol/L].

Study Design

It's an open-label, prospective, observational, multinational, registry-based population cohort design.

Participants → **CGM/BGM groups** → **Follow-up** → **Outcomes**

Size: 339

Age: 1-7 years old

Inclusion Criteria:

1. T1D diagnosed at least 6 months prior
2. Insulin pump use for at least 3 months

CGM

VS

Finger stick blood glucose monitoring alone (**BGM**)

Duration: 12 months

CGM group:

≥10 days of data every 3 months

BGM group:

≥10 days of data with a minimum of five data points per day every 3 months

Glycemic metrics:

HbA1c

TIR

Time above range (TAR)

Time below range (TBR)

Coefficient of variation (CV)

Other

TIR: 70-180 mg/dl [3.9-10.0 mmol/L], TBR: <70 mg/dl) [<3.9 mmol/L]), TAR: >180 mg/dl [>10 mmol/L].

HbA1c was measured in each centre and standardized to the Diabetes Control and Complications Trial reference of 20-42 mmol/mol (4%-6%).

The HbA1c target: <7% (<53 mmol/mol) for most children with T1D, <6.5% (<48 mmol/mol) for more stringent goals.

Endpoints

Primary Endpoint:

A difference in glycemic variability, measured by CV, between the CGM and BGM groups.

Other Endpoints:

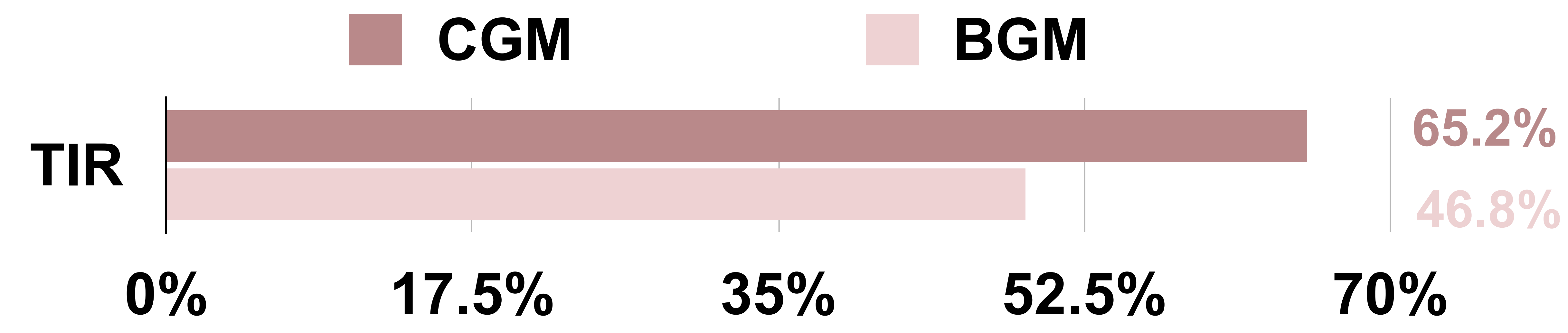
Recommended standardized glycemia metrics (including TIR, TAR, TBR and HbA1c)

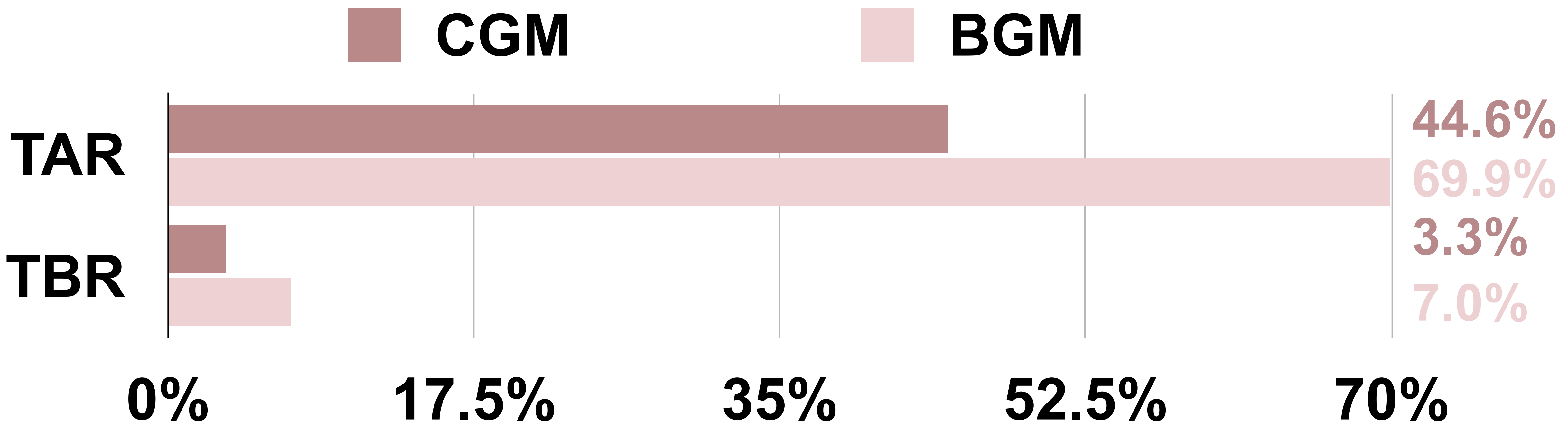
Primary Result

The CV among CGM and BGM users were 39.1% and 46.8%, respectively.

➡ Indicated smaller glycemic variability in the CGM group than the BGM group.

Other Results





1. The proportion of TIR was about 15% more in the CGM than the BGM group, with less pronounced excursions in the ambulatory glucose profile for the former.

2. The CGM group spent less time above and below TIR than the BGM group.

3. The median HbA1c was 7.0% and 7.5% for the CGM and BGM groups, respectively.

4. 33.7% (59 users) of the CGM achieved the clinical target of TIR more than 70%, comparing to the 3.8% (2) for the BGM.

5. One severe hypoglycemia event occurred in the CGM and four among the BGM cohort. Diabetic ketoacidosis occurred twice in one CGM user.

*The results in the bar charts were measured over the full 24-hour period

Clinical Relevance

CGM use being associated to both diminished glucose variability and reduced hypoglycemia,

even while maintaining HbA1c,

could have long-term clinical benefit for very young children with T1D.

hypoglycaemia: <70 mg/dL.

Strengths & Limitations

Strengths:

1. The largest prospective multinational real-world analysis of glycemic metrics beyond HbA1c in very young children with T1D using CGM to date.
2. Population heterogeneity
   - 15 centers in 9 countries
   - Wide glycemia baseline, with HbA1c ranging from 5.5% to 10%.

Limitations:

1. Open-label design may lead to skewed results.
2. Included only those on pump therapy, cannot extrapolate to those on multiple daily injections.
3. Didn’t systematically record the use of suspend at low or predictive low suspend functions in the CGM.
   - The semi-automated insulin therapy was not yet approved in this age range.
4. No socioeconomic and psychological data.
   - 15 centers from a variety of countries and systems providing healthcare, multiple factors may have impacted participants’ use of CGM and/or pump technology, including availability, access, provider assessment, and family preference.

Conclusions

1. CGM use is associated with reduced glucose fluctuations in real-world data for very young individuals with T1D.

2. Many young children on pumps who were also using CGM approached targeted glycemia, as measured by both HbA1c and TIR.

For future study

Additional long-term studies are needed to determine if CGM, especially when incorporated into semi-automated insulin delivery, will improve outcomes, including:

- Daily glycemia, rates of both acute and chronic complications, and diabetes care-related burden.

Supplemental materials

Baseline characteristics of study participants

	All (n = 227)	CGM (n = 175)	BGM only (n = 52)
Age (y)	5.3 (4.1, 6.4)	5.2 (4.0, 6.4)	5.5 (4.7, 6.3)
Age at diagnosis (y)	2.5 (1.6, 3.9)	2.5 (1.6, 3.8)	2.6 (1.6, 3.9)
Duration of diabetes (y)	2.1 (1.4, 3.3)	2.2 (1.5, 3.2)	2.2 (1.2, 3.4)
Body weight (kg)	20.0 (17.2, 23.0)	19.6 (17.1, 22.3)	21.8 (18.2, 23.8)
Body height (cm)	112 (104, 121)	111 104, 119)	115 (108, 121)
BMI	16.1 (15.2, 17.1)	16.2 (15.2, 17.0)	16.1 (15.5, 17.1)
BMI SDS	0.7 (0.0, 1.4)	0.7 (0.0, 1.2)	0.6 (−0.1, 1.8)
Female gender, n (%)	95 (42)	78 (45)	17 (33)
Non-Hispanic White (%)	195 (86)	147 (84)	50 (96)
HbA1c (%)	7.2 (6.6, 7.8)	7.0 (6.5, 7.6)[a]	7.6 (7.1, 8.1)[a]
HbA1c (mmol/mol)	55 (49, 62)	53 (48, 60)[a]	60 (54, 65)[a]

Data are median (IQR) or count (%).

BMI, body mass index; **BMI SDS**, Body Mass Index Standard Deviation Score, calculated based on World Health Organization references.

[a] There were no statistically significant differences in baseline characteristics between study groups, except for HbA1c (P = .007).